

Assignment for DAT246/DIT246

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CAUSAL ASSUMPTIONS & DAGs

We study the effect of `used.cr.technology` on `result` in a code-review setting. Review `extent` and review `quality` are mediators. Reviewer `age`, `skill`, and code `complexity` are background variables.

Figure 1 shows the first causal model. In this DAG, `age` affects `skill`, which in turn affects `extent` and `quality`. `Age` also affects `used.cr.technology`, and `complexity` affects `extent` and `quality`. `used.cr.technology` then influences `extent`, `quality`, and directly `result`. Finally, `extent` and `quality` affect `result`. The *total effect* of `used.cr.technology` on `result` is defined along all directed paths from `used.cr.technology` to `result` and is identified by adjusting for the pre-treatment confounder `age` only. We do not condition on mediators `extent` or `quality` when estimating this total effect.

Total effect of A on R without conditioning on mediators E and Q.

Direct effect of A on result conditioning on mediators - E and Q, and confounders - G, S, and C.

In the second model, Figure 2, we make the adjustment strategy explicit. We distinguish three estimands. (i) The *total effect* of A on R without conditioning on E and Q. (ii) The *adjusted effect* controlling for confounders G, S, and C but not for mediators, to reduce variance while preserving the total effect. (iii) The *direct effect* of A on R, conditional on mediators E and Q and confounders G, S, and C, which blocks the indirect paths through E and Q. These adjustment sets are consistent with the back-door criterion under the assumed DAGs.

In this model, A affects outcomes mainly via S and C:

Total effect of A on R without conditioning on mediators E and Q.

Adjusted effect of A on R conditioning on confounders G, S, and C but not mediators E and Q.

Direct effect of A on R conditioning on mediators E and Q and confounders G, S, and C.

DESCRIPTIVE STATISTICS & VISUALS

Figure 3 shows the distribution of AI usage. This is a descriptive check of *exposure balance*. The dataset contains a comparable number of AI-assisted (1) and non-AI (0) reviews. Such balance is important for causal inference, since estimating the effect of `used.cr.technology` on `result` requires sufficient variation in the exposure.

Figure 4 presents the distribution of approval outcomes. This is a de-

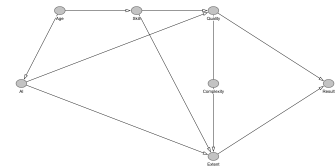


Figure 1: DAG 1: Affects of age, skill, complexity, and AI usage on review extent, quality, and result.

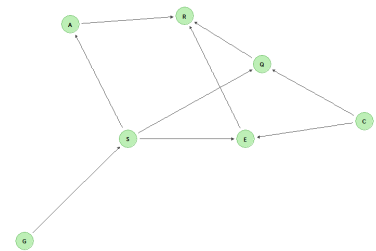
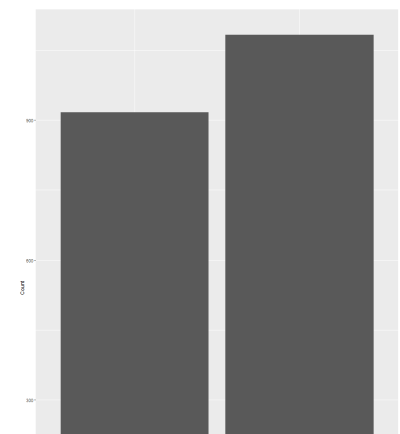


Figure 2: Dag 2



scriptive check of *outcome balance*. Approved (1) and not-approved (0) reviews appear in similar numbers, indicating that the logistic regression is well-supported. The approval process is not degenerate; reviewers frequently reject or flag reviews.

Figure 5 displays a scatter plot of standardized complexity against standardized extent. This acts as a pre-model diagnostic check. The strong positive linear trend indicates that more complex code changes are associated with longer or more extensive reviews. This supports the DAG assumption $complexity \rightarrow extent$ and suggests that *complexity* will have a positive slope in the extent model.

Figure 6 shows the relationship between standardized skill and standardized review quality. The data appear clustered in bands—likely due to quality being discretized—but the fitted line still exhibits a positive trend. Higher-skill reviewers tend to write higher-quality reviews. This supports the DAG assumption $skill \rightarrow quality$ and justifies including *skill* as a predictor in the GLM for review quality.

STATISTICAL MODELS

We design three statistical models corresponding to the three outcomes *extent*, *quality*, and *result*. The choice of predictors follows directly from our causal assumptions encoded in the Figure 1, and the likelihoods follow from the maximum entropy principle and generalized linear model theory.

Model for *extent*:

According to the DAG, the direct parents of *extent* are *skill*, *complexity*, and *used.cr.technology*. These variables exert causal influence on review length: skilled reviewers may produce shorter or more focused reviews, complex changes require longer reviews, and AI usage may alter the structure or size of the written review. *Age* affects *extent* only indirectly via *skill* and AI usage. It is therefore not required for causal identification but may serve as a precision variable. Because *extent* is continuous and constrained only by its mean and variance, the maximum entropy distribution is Gaussian. We therefore model

$$extent_i \sim \text{Normal}(\mu_i, \sigma), \quad \mu_i = \alpha + \beta_{\text{skill}} \text{skill}_i + \beta_{\text{complexity}} \text{complexity}_i + \beta_{\text{used}} \text{used}_i.$$

Model for *quality*:

The DAG indicates the same parent set for *quality*: *skill*, *complexity*, and *used.cr.technology*. Skilled reviewers are expected to write clearer reviews, complex changes lower review quality, and AI usage may affect wording or clarity. As with the previous model, *age* is upstream but not a direct cause. *Quality* is treated as a continuous score, and with only mean and variance as constraints, the maximum entropy choice is again the Gaussian distribution:

$$quality_i \sim \text{Normal}(\mu_i, \sigma), \quad \mu_i = \alpha + \beta_{\text{skill}} \text{skill}_i + \beta_{\text{complexity}} \text{complexity}_i + \beta_{\text{used}} \text{used}_i.$$

Model for *Result* (Total Effect):

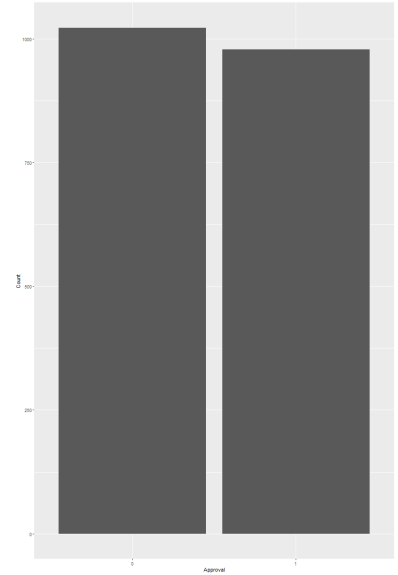
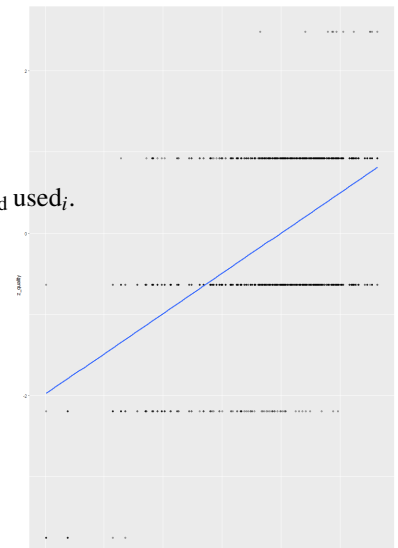


Figure 4: Distribution of approval outcomes.



Figure 5: Complexity vs. review extent.



To estimate the *total* causal effect of AI usage on approval, we include only the exposure *used.cr.technology* and adjust for the confounder *age*. The DAG shows a backdoor path $\text{used} \leftarrow \text{age} \rightarrow \text{skill} \rightarrow \{\text{extent}, \text{quality}\} \rightarrow \text{result}$, making *age* necessary for identification. We do not condition on *extent* or *quality*, since they are mediators. *Result* is binary, so the maximum entropy distribution is Binomial with a logit link:

$$\text{result}_i \sim \text{Binomial}(1, p_i), \quad \text{logit}(p_i) = \alpha + \beta_{\text{used}} \text{used}_i + \beta_{\text{age}} \text{age}_i.$$

Model for Result (Direct Effect):

To estimate the *direct* effect of AI usage on approval, we must block the indirect paths through the mediators *extent* and *quality*. Thus, we condition on these mediators and their other parents (*skill*, *complexity*, and *age*) to avoid introducing collider bias. The resulting model is

$$\text{logit}(p_i) = \alpha + \beta_{\text{used}} \text{used}_i + \beta_{\text{age}} \text{age}_i + \beta_{\text{skill}} \text{skill}_i + \beta_{\text{complexity}} \text{complexity}_i + \beta_{\text{extent}} \text{extent}_i + \beta_{\text{quality}} \text{quality}_i.$$

This specification isolates the direct effect of AI usage by conditioning on the mediator values and blocking all indirect causal pathways.

SENSIBLE PRIORS

All continuous predictors and outcomes are standardized to mean zero and unit variance. This allows us to use weakly informative but regularizing Gaussian priors on regression coefficients. For all intercepts we use

$$\alpha. \sim \text{Normal}(0, 1),$$

and for all slopes we use

$$\beta_{.,.} \sim \text{Normal}(0, 0.5).$$

These priors encode that most effects are likely modest on the standardized scale, while still allowing larger effects if supported by the data. For Gaussian outcome standard deviations we use

$$\sigma_E, \sigma_Q \sim \text{Exponential}(1),$$

which constrains them to be positive and mildly shrinks them towards one. These choices follow the maximum-entropy justification of common likelihoods and provide regularization that improves out-of-sample performance.

CONCLUSION

The DAGs specify clear causal assumptions about how AI usage, reviewer characteristics, and change complexity jointly influence review extent, quality, and approval. Descriptive statistics and visuals support key edges in the DAG, such as *complexity* $\rightarrow$ *extent* and *skill* $\rightarrow$ *quality*. The GLM formulations with weakly informative priors are consistent with these assumptions

and stable under MCMC sampling. Using PSIS-LOO for model comparison, we can distinguish between models targeting the total and direct effects of AI usage on approval. Overall, the analysis suggests that AI usage has both direct and indirect effects on approval, mediated by changes in review extent and quality, while accounting for age, skill, and complexity as confounding factors.

```
d <- read.csv("foo.csv")  
str(d)
```